

A Framework for Efficient Search of High-Distance Quantum Stabilizer Codes

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Abstract

Quantum stabilizer codes are essential for protecting quantum information, but systematic searches for high-distance codes become computationally prohibitive as code length increases. Existing code constructions exhibit structural regularities, yet no general framework quantifies how such constraints reduce the stabilizer-code search space or incorporates them into code searches. This project asks whether structural properties of known quantum codes can be formulated as mathematical constraints that quantitatively reduce search spaces while remaining useful for discovering high-distance codes. We develop a unified framework based on q -Galois matrices and define the search-space retention factor as the asymptotic fraction of logarithmic search space retained after imposing a constraint. We derive retention factors for weight-spectrum, qudit-participation, overlap, support, Pauli-distribution, local non-commutation, graph-theoretic, and symmetry constraints, including combinations. We introduce finite-size correction factors C and C' and find empirically that they decay approximately as $\log n/n$ and $1/n$, respectively. The framework is evaluated using 58,902 quantum-code parameter sets, including 34,775 codes with usable matrix data across $q = 2, 3, 4, 5, 7, 8$; 1,909 semi-perfect quantum codes are analyzed. We develop a general method to empirically estimate the weight-spectrum and average-weight for unresolved semi-perfect quantum codes, achieving runtime speedups ranging from $10^{8.3608}$ to $10^{193.9298}$ across 4 distinct unresolved codes. Runtime analysis validates that these reductions translate into substantial computational savings. Applying the framework to a $(14, 3)$ qubit search produced a code detecting 91,765 of 91,770 errors of weight at most 4 and achieving 99.99999933% and 99.9733% of the theoretically optimal performance of $[[14, 3, 5]]_2$ code at $p = 0.01$. Thus, the framework converts structural properties into explicit candidate-space reductions that can be imposed during code enumeration, providing a quantitative basis for reducing the computational cost of high-distance stabilizer-code searches.

Keywords: Quantum error correction, quantum stabilizer codes, search-space retention factor, quantum code search, finite-size corrections, semi-perfect quantum codes, quantum code construction

1 Introduction

Quantum error-correcting codes provide a means of protecting quantum information against noise and physical errors [1]. Among them, quantum stabilizer codes are particularly important due to their rich algebraic structure and efficient representations. However, systematic searches for high-distance stabilizer codes become computationally prohibitive as the code length increases, motivating methods that exploit structural properties of known codes to reduce the number of candidate configurations that must be examined.

In this work, we introduce a unified mathematical framework for incorporating structural constraints into systematic searches for high-distance quantum stabilizer codes, supported by extensive empirical analysis and computational validation. We represent stabilizer generators using q -Galois matrices and introduce the qudit participation function, local non-commutation count, and search-space retention factor to quantify structural properties and their effect on the search space. We further define a family of semi-perfect quantum codes, which forms the central family for our empirical analysis. Using these quantities, we formulate and quantify the effectiveness of weight-spectrum, qudit-participation, overlap, support, Pauli-distribution, local non-commutation, graph-theoretic, symmetry, and mixed constraints.

We derive a finite-size correction factor to account for deviations between finite-length search spaces and their asymptotic retention factors. The correction is empirically analyzed using 58,902 quantum-code parameter sets, including 34,775 quantum codes with usable matrix data across $q = 2, 3, 4, 5, 7$, and 8, including 1,909 semi-perfect quantum codes. We further develop a general method for empirically estimating the weight spectrum and average weight of semi-perfect quantum codes. Applying these estimates to four distinct unresolved semi-perfect codes yields runtime speedups ranging from $10^{8.3608}$ to $10^{193.9298}$.

We computationally validate the framework through runtime analysis on the $[[5, 1, 3]]_2$ code, demonstrating that the predicted reductions in search space translate into practical reductions in computational cost. The framework is independent of the underlying search algorithm and can therefore be incorporated into different systematic and constraint-based search procedures.

Finally, we apply the framework to a $(14, 3)$ qubit search and obtain a high-performance code that detects 91,765 of 91,770 errors of weight at most 4. Under a physical error probability of $p = 0.01$, the resulting code achieves 99.99999933% and 99.9733% of the theoretical performance of the optimal $[[14, 3, 5]]_2$ code, respectively, demonstrating the practical utility of the framework in an actual high-distance code search.

2 Preliminaries

2.1 Notations

Throughout this paper, the following notation will be used.

- S : the stabilizer group.
- $\mathcal{G}$: a generating set of the stabilizer group S .
- $g \in \mathcal{G}$: a stabilizer generator.
- $\text{wt}(g)$: the weight of the stabilizer generator g .
- $\text{supp}(g)$: the support of g , i.e., the set of qudits on which g acts non-trivially.
- $N_X(g_i)$, $N_Y(g_i)$, $N_Z(g_i)$: the number of Pauli- X , Pauli- Y , and Pauli- Z operators, respectively, appearing in the stabilizer generator g_i .

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$$N_X = \sum_{i=1}^{n-k} N_X(g_i), \quad N_Y = \sum_{i=1}^{n-k} N_Y(g_i), \quad N_Z = \sum_{i=1}^{n-k} N_Z(g_i),$$

denote the total numbers of Pauli- X , Pauli- Y , and Pauli- Z operators appearing in the generating set.

- O : the Gram matrix corresponding to the stabilizer generators.
- $H_q(p)$: the q -ary entropy function: $H_q(p) = p \log_q(q-1) - p \log_q p - (1-p) \log_q(1-p)$

2.2 Definitions

Semi-Perfect Quantum Code

Let $[[n, k, d]]_q$ denote a non-degenerate quantum code. The code is called *semi-perfect* if its parameters lie on the boundary of the currently known physically allowable parameter space determined by the quantum Hamming bound, the Rains bound [2], the quantum Singleton bound and other known non-existence results.

Equivalently, each parameter is individually optimal while the remaining two are held fixed. That is,

- there exists no code $[[n', k, d]]_q$ with $n' < n$;
- there exists no code $[[n, k, d']]_q$ with $d' > d$; and
- there exists no code $[[n, k', d]]_q$ with $k' > k$.

Thus, a semi-perfect quantum code is simultaneously optimal with respect to block length, distance, and encoded dimension, subject to the remaining two parameters being fixed. Every perfect quantum code is semi-perfect, although a semi-perfect quantum code need not be perfect.

q -Galois String

A q -Galois string represents a stabilizer generator for a q -ary quantum stabilizer code. It is defined as a length- n string:

$$\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n), \quad \alpha_i \in \text{GF}(q^2).$$

More generally, a q -Galois string provides a representation of any n -qudit Pauli operator, with stabilizer generators corresponding to the subset of such operators that generate the stabilizer group.

By fixing a basis of $\text{GF}(q^2)$ over $\text{GF}(q)$, each element can be expressed as:

$$\alpha_i = a_i + \omega b_i, \quad a_i, b_i \in \text{GF}(q).$$

Up to a phase convention, these components map directly to the generalized Pauli operator $X^{a_i} Z^{b_i}$. Thus, the complete q -Galois string corresponds to the tensor product of these operators, representing the full stabilizer generator:

$$\alpha \longleftrightarrow \bigotimes_{i=1}^n X^{a_i} Z^{b_i}.$$

Unless stated otherwise, a 2-Galois string is referred to simply as a "Galois string." In this standard scenario, the string is defined over $\text{GF}(4)$ [3].

q -Galois Matrix

We define the q -Galois matrix, denoted by G , as the $(n - k) \times n$ matrix whose i th row is the q -Galois string corresponding to the i th stabilizer generator. That is,

$$G = \begin{bmatrix} \alpha^{(1)} \\ \alpha^{(2)} \\ \vdots \\ \alpha^{(n-k)} \end{bmatrix},$$

or equivalently,

$$G_{ij} = \alpha_j^{(i)},$$

where $1 \leq i \leq n - k$ and $1 \leq j \leq n$.

Thus, each row of G represents a stabilizer generator in its q -Galois string representation, while each column corresponds to a physical qudit.

Local Non-Commutation Count

For two n -qudit Pauli operators P_1 and P_2 , we define the *local non-commutation count*, denoted by $C(P_1, P_2)$, as the number of qudits on which the corresponding single-qudit Pauli operators don't commute.

Let α_1 and α_2 denote the q -Galois strings corresponding to P_1 and P_2 , respectively. Then

$$C(P_1, P_2) = \sum_{i=1}^n \chi((\alpha_1)_i, (\alpha_2)_i),$$

where

$$\chi(a, b) = \begin{cases} 0, & (a + b)ab = 0, \\ 1, & (a + b)ab \neq 0. \end{cases}$$

The summation is performed over the integers; in particular, $C(P_1, P_2)$ is not evaluated modulo two.

For stabilizer generators g_i and g_j , we write

$$C_{ij} = C(g_i, g_j).$$

Qudit Participation Function

For any matrix M , we define the *qudit participation function* on its columns by

$$Q(M|a) = \sum_i \Lambda((M|a)_i),$$

where $M|a$ denotes the $(a + 1)$ th column of M , and

$$\Lambda(x) = \begin{cases} 0, & x = 0, \\ 1, & x \neq 0. \end{cases}$$

Thus, $Q(M|a)$ counts the number of nonzero entries in the $(a + 1)$ th column of M . In particular, $Q(G|a)$ denotes the number of stabilizer generators acting non-trivially on the $(a + 1)$ th physical qudit.

p -wise Overlap

Let $g_{i_1}, g_{i_2}, \dots, g_{i_p}$ be p stabilizer generators. We define the *p -wise overlap* of these generators as

$$I_p(i_1, i_2, \dots, i_p) = |\text{supp}(g_{i_1}) \cap \text{supp}(g_{i_2}) \cap \dots \cap \text{supp}(g_{i_p})|.$$

Equivalently, $I_p(i_1, i_2, \dots, i_p)$ denotes the cardinality of the common support shared by the selected stabilizer generators.

Stabilizer Overlap Graph

Let τ be a prescribed positive integer. We define the *Stabilizer Overlap Graph* (SOG) with threshold τ , denoted by $\text{SOG}(\tau)$, as the graph whose vertices correspond to the stabilizer generators. Two distinct vertices corresponding to generators g_i and g_j are connected by an edge if and only if

$$|\text{supp}(g_i) \cap \text{supp}(g_j)| \geq \tau.$$

The parameter τ is called the *threshold* of the graph.

Unless stated otherwise, all graph-theoretic terminology and notation follow their standard definitions.

Search Space Retention Factor

For a constraint imposed on a family of codes $[[n, k_n, d]]_q$, we define the search space retention factor by

$$\mathcal{R}_{[[n, k_n, d]]_q} = \lim_{n \rightarrow \infty} \frac{\log_{q^2} S_R}{\log_{q^2} S_O},$$

where S_O and S_R denote the sizes of the original and reduced search spaces, respectively.

The *search-space retention factor* provides a measure of the fraction of the logarithmic search space retained after imposing a constraint.

Equivalently, for a single finite code, we define the search-space retention factor $\mathcal{R}$ as

$$\mathcal{R} = \frac{\log_{q^2} S_R}{\log_{q^2} S_O} = \mathcal{R}_{[[n, k_n, d]]_q} - C,$$

where C is a finite-size correction factor accounting for the finiteness of n . The correction factor vanishes asymptotically, satisfying

$$\lim_{n \rightarrow \infty} C = 0.$$

The search-space size may be interpreted as the number of possible configurations of the q -Galois matrix, or equivalently the corresponding parity-check representation. By default, we take

$$S_O = q^{2n(n-k_n)},$$

corresponding to an unrestricted q -Galois matrix, with each entry independently taking any value in $\text{GF}(q^2)$.

A smaller search-space retention factor indicates a stronger reduction of the search space, but may also increase the risk of excluding the desired code. Conversely, a larger search-space retention factor corresponds to a weaker restriction and a lower risk of excluding valid constructions. The objective is therefore to identify constraints that minimize the search-space retention factor while retaining the desired code, with the appropriate choice depending on the target family and its neighboring code parameters.

For the universal stabilizer-code constraints

$$\text{rank}(H) = n - k_n, \quad H_X H_Z^T = H_Z H_X^T,$$

the corresponding reduced search-space size is

$$S_R = \prod_{i=1}^{n-k_n} (q^{2n-i+1} - q^{i-1}).$$

Consequently, the search-space retention factor is

$$\mathcal{R}_{[[n,k_n,d]]_q} = \frac{3+r}{4}.$$

where

$$r = \lim_{n \rightarrow \infty} \frac{k_n}{n}$$

is the asymptotic rate of the code family.

We disregard this search-space retention factor in the subsequent analysis, since these constraints are common to all stabilizer codes and can be imposed directly during computation. Consequently, they only reduce or leave unchanged the search space available to the additional structural constraints considered below.

3 Structural Assumptions

This section develops the structural assumptions considered in the proposed framework, including weight, participation, overlap, graph-theoretic, support, Pauli-distribution, local anticommutation, and symmetry-based constraints.

3.1 Weight Spectrum

One possible structural restriction is to require every stabilizer generator to have weight belonging to a prescribed weight spectrum.

Let

$$\mathcal{W} = \{w_1, w_2, \dots, w_m\}, \quad 1 \leq w_i \leq n, \quad w_i \in \mathbb{Z}_{>0}, \quad \text{wt}(g) \in \mathcal{W} \quad \forall g \in \mathcal{G}.$$

The constant-weight case is recovered when $|\mathcal{W}| = 1$.

Correspondingly,

$$S_R = \left(\sum_{i=1}^m \binom{n}{w_i} (q^2 - 1)^{w_i} \right)^{n-k_n}, \quad \mathcal{R}_{[[n,k_n,d]]_q} = H_{q^2}(\delta'),$$

where

$$\delta = \lim_{n \rightarrow \infty} \frac{\max(\mathcal{W})}{n}.$$

Examples of semi-perfect quantum codes and their corresponding weight spectra include: $[[5, 1, 3]]$ [4]: $\mathcal{W} = \{4\}$; $[[11, 1, 5]]$ [4]: $\mathcal{W} = \{6, 8\}$; and $[[n, n-2, 2]]$ [5]: $\mathcal{W} = \{n\}$, for every even integer $n \geq 2$.

3.2 Qudit Participation

3.2.1 Regular Qudit Participation

Another natural assumption is that every physical qudit participates in the same number of stabilizer generators. Let w denote the average stabilizer-generator weight and r the constant qudit participation. Then

$$Q(G|a\rangle) = r, \quad \forall a \in \{0, 1, \dots, n-1\}; \quad (n-k)w = nr, \quad w = \frac{\sum_i \text{wt}(g_i)}{n-k}.$$

Correspondingly,

$$S_R = \left[\binom{n-k_n}{r} (q^2 - 1)^r \right]^n, \quad \mathcal{R}_{[[n, k_n, d]]_q} = H_{q^2}(\delta'),$$

where

$$\delta' = \lim_{n \rightarrow \infty} \frac{r}{n - k_n}.$$

3.2.2 Restricted Qudit Participation

One possible structural restriction is to impose an upper bound on the participation of each physical qudit. We require

$$Q(G|a) \leq r_{\max}, \quad \forall a \in \{0, 1, \dots, n-1\}.$$

Thus, no physical qudit participates in more than $r_{\max}$ stabilizer generators, reducing the combinatorial search space while retaining flexibility in code construction.

Correspondingly,

$$S_R = \left(\sum_{i=0}^{r_{\max}} \binom{n-k_n}{i} (q^2 - 1)^i \right)^n, \quad \mathcal{R}_{[[n, k_n, d]]_q} = H_{q^2}(\delta'),$$

where

$$\delta' = \lim_{n \rightarrow \infty} \frac{r_{\max}}{n - k_n}.$$

3.3 Overlap Constraints

3.3.1 Restrictions on Gram Matrix

(a) Overlap-Value Constraints

A natural class of structural restrictions is to constrain the overlap values encoded in the Gram matrix. Three possible constraints are:

- **Symmetric Design:** require every pair of distinct stabilizer generators to have the same overlap,

$$O_{ij} = \begin{cases} \text{wt}(g_i), & i = j, \\ \lambda, & i \neq j, \end{cases}$$

where λ is a fixed non-negative integer. An example of a semi-perfect quantum code exhibiting this property is: $[[5, 1, 3]]$: $\lambda = 3$.

- **Quasi-Symmetric Design:** allow exactly two possible overlap values between distinct stabilizer generators,

$$O_{ij} = \begin{cases} \text{wt}(g_i), & i = j, \\ \lambda \text{ or } \mu, & i \neq j, \end{cases}$$

where λ and μ are fixed non-negative integers. Examples of semi-perfect quantum codes that exhibit this property include: $[[8, 3, 3]]$ [4], $[[12, 4, 4]]$ [6], and $[[12, 6, 3]]$ [6].

- **Bounded Overlap:** impose an upper bound on the overlap between distinct stabilizer generators,

$$O_{ij} \leq \lambda, \quad \forall i \neq j.$$

The corresponding search-space retention factors for the three constraints above are, respectively,

$$\begin{aligned}\mathcal{R}_{[[n,k_n,d]]_q} &= \sqrt{\delta} \log_{q^2}(q^2 - 1), & \mathcal{R}_{[[n,k_n,d]]_q} &= \sqrt{\delta_{max}} \log_{q^2}(q^2 - 1), \\ \mathcal{R}_{[[n,k_n,d]]_q} &= \frac{H_q(\sqrt{\delta}) + \sqrt{\delta} \log_q(q+1)}{2} & \text{for } \sqrt{\delta} &\leq 1 - \frac{1}{q^2}, \\ \mathcal{R}_{[[n,k_n,d]]_q} &= 1 & \text{for } \sqrt{\delta} &\geq 1 - \frac{1}{q^2}.\end{aligned}$$

where

$$\delta = \lim_{n \rightarrow \infty} \frac{\lambda}{n} \quad \delta_{max} = \max\left(\lim_{n \rightarrow \infty} \frac{\lambda}{n}, \lim_{n \rightarrow \infty} \frac{\mu}{n}\right).$$

(b) Eigenvalue Constraints

Instead of restricting individual entries of the Gram matrix, one may constrain its spectrum by requiring it to have only two distinct eigenvalues, E_1 and E_2 .

An example of a semi-perfect quantum code exhibiting this property is: $[[5, 1, 3]]$: $E_1 = 1$, $E_2 = 13$.

3.3.2 General p -wise Overlap Constraints

A natural generalization of pairwise overlap is to impose regularity on the common overlap of p distinct stabilizer generators. Three possible constraints are:

- **Symmetric p -wise Overlap:** Require every collection of p distinct stabilizer generators to have the same common overlap:

$$I_p(i_1, \dots, i_p) = \lambda_p,$$

for every choice of distinct indices $i_1, \dots, i_p$, where λ_p is a constant depending only on p .

- **Quasi-Symmetric p -wise Overlap:** Allow exactly two possible overlap values:

$$I_p(i_1, \dots, i_p) \in \{\lambda_p, \mu_p\},$$

where $\lambda_p \neq \mu_p$.

- **Bounded p -wise Overlap:** Rather than restricting the overlaps to exact values, require

$$I_p(i_1, \dots, i_p) \leq \lambda_p, \quad \forall i_1, \dots, i_p \text{ distinct}.$$

The corresponding search-space retention factors for the three constraints above are, respectively,

$$\begin{aligned}\mathcal{R}_{[[n,k_n,d]]_q} &= \delta^{1/p} \log_{q^2}(q^2 - 1), & \mathcal{R}_{[[n,k_n,d]]_q} &= \delta_{max}^{1/p} \log_{q^2}(q^2 - 1), \\ \mathcal{R}_{[[n,k_n,d]]_q} &= \frac{H_q(\delta^{1/p}) + \delta^{1/p} \log_q(q+1)}{2} & \text{for } \delta^{1/p} &\leq 1 - \frac{1}{q^2}, \\ \mathcal{R}_{[[n,k_n,d]]_q} &= 1 & \text{for } \delta^{1/p} &\geq 1 - \frac{1}{q^2}.\end{aligned}$$

where

$$\delta = \lim_{n \rightarrow \infty} \frac{\lambda}{n} \quad \delta_{max} = \max\left(\lim_{n \rightarrow \infty} \frac{\lambda}{n}, \lim_{n \rightarrow \infty} \frac{\mu}{n}\right).$$

3.3.3 Structural Constraints on SOG

Before imposing structural constraints on the SOG, note that their search-space retention factor is generally case-dependent. Such constraints may be analyzed through combinatorial structures such as balanced incomplete block designs, as SOG properties can be formulated in terms of stabilizer-support intersections and expressed through the corresponding Gram and Galois matrices. We therefore assign no universal retention factor to SOG constraints. The SOG is used primarily to identify structural patterns in known semi-perfect quantum codes rather than as a direct computational search representation.

(a) Constraints on SOG Parameters

One may impose upper bounds on the clique number, independence number, degree, and diameter of the SOG, or a lower bound on its girth.

These constraints control different aspects of the overlap structure: a lower girth suppresses short cycles and localized overlap patterns, while upper bounds on the clique number, independence number, and degree limit excessive mutual overlap, weakly interacting generator sets, and highly connected “hub” generators, respectively. An upper bound on the diameter promotes a highly interconnected overlap structure in which stabilizer generators are separated by only a small number of graph edges.

(b) Constraints on the Eigenvalues of the Adjacency Matrix

Rather than restricting the combinatorial properties of the SOG directly, one may constrain the spectrum of its adjacency matrix. Its eigenvalues capture important global structural properties of the graph, with highly symmetric graphs often having only a few distinct eigenvalues. Accordingly, one may require the adjacency matrix of the SOG to have only a small number of distinct eigenvalues, potentially enforcing hidden combinatorial symmetries while reducing the search space.

(c) Regularity Conditions

Another possible restriction is to require the SOG to satisfy additional graph-theoretic regularity conditions. For example, the SOG may be required to be regular, strongly regular, or distance-regular [7]. Such restrictions impose increasing levels of structural symmetry on the overlap graph, thereby reducing the number of admissible candidate configurations and potentially simplifying the search for semi-perfect quantum codes.

(d) Chordal SOG

A chordal SOG eliminates large chordless cycles and imposes a more hierarchical graph structure, potentially simplifying both the combinatorial analysis and the search for candidate codes. Chordal structure appears to be highly common among semi-perfect quantum codes, particularly those with fewer encoded qudits.

(e) Vertex Transitivity

Another structural restriction is requiring the SOG to be vertex-transitive, meaning that any pair of vertices can be mapped to each other by a graph automorphism. Thus, all stabilizer generators occupy equivalent structural roles, eliminating structurally equivalent cases from the search.

(f) Edge Transitivity

A further structural restriction is to require the SOG to be edge-transitive, meaning that any pair of edges can be mapped to each other by a graph automorphism. Thus, all overlap relations between stabilizer generators are structurally equivalent, eliminating redundant cases from the search.

3.4 Non-Contained Supports

One possible structural restriction is to require that no stabilizer support be a subset of another:

$$\text{supp}(g_i) \not\subseteq \text{supp}(g_j), \quad \forall i \neq j.$$

Equivalently, every pair of stabilizer generators has at least one qudit on which one acts non-trivially while the other does not. For this constraint, the search-space retention factor is

$$\mathcal{R}_{[[n, k_n, d]]_q} = 1.$$

This indicates that the constraint provides little reduction of the search space when considered independently. Nevertheless, it may be partially incorporated alongside other structural constraints by identifying patterns across a family of codes, potentially making it useful in combination with stronger restrictions, particularly for small values of n . Although it appears to be satisfied by the $[[5, 1, 3]]$ and $[[8, 3, 3]]$ semi-perfect quantum codes, its effectiveness as an independent search constraint remains limited.

3.5 Pauli Distribution Constraints

The Pauli operators X , Y , and Z play symmetric roles in quantum mechanics, motivating restrictions on their distribution within a stabilizer generating set. However, such constraints are generally not invariant under local Clifford transformations and therefore apply to a particular choice of generators rather than the underlying code. Nevertheless, they may serve as useful search heuristics when identifying a representative satisfying the desired code parameters.

The possible constraints considered are:

- **Generator-Wise Equal Distribution:** Require

$$N_X(g_i) = N_Y(g_i) = N_Z(g_i), \quad \forall i.$$

This requires every generator weight to be divisible by three.

- **Globally Equal Distribution:** Require equality only across the complete generating set:

$$N_X = N_Y = N_Z.$$

- **Quasi-Equal Distribution:** Allow deviations of at most one, either generator-wise or globally:

$$|N_X - N_Y|, |N_Y - N_Z|, |N_Z - N_X| \leq 1.$$

- **Restriction on Variance:** Require the Pauli distribution to have bounded variance:

$$\text{Var}(N_X, N_Y, N_Z) \leq \sigma^2.$$

- **Semi-Equal Distribution:** Require two Pauli types to occur equally while allowing the third to be absent, e.g.,

$$N_X = N_Z, \quad N_Y = 0.$$

An example of a semi-perfect quantum code exhibiting this property is: $[[5, 1, 3]]$: $N_Y = 0$.

The constraints above are stated generally for qudit stabilizer codes, but are illustrated explicitly in the qubit case.

For the generator-wise equal distribution, globally equal distribution, quasi-equal distribution, and restriction on variance constraints, the search-space retention factor is

$$\mathcal{R}_{[[n, k_n, d]]_q} = 1.$$

For the semi-equal distribution constraint, the search-space retention factor must instead be evaluated case by case, depending on the number and choice of Pauli operators required to have equal global occurrence.

We get $\mathcal{R}_{[[n,k_n,d]]_q} = 1$ because, as $n \rightarrow \infty$, the random assignment of Pauli operators to the entries of the Galois matrix naturally approaches an equal distribution among the Pauli operators. Consequently, these constraints do not alter the leading exponential order of the search-space size. This provides a further mathematical indication of the equivalent roles played by X , Y , and Z in quantum mechanics.

3.6 Local Non-Commutation Constraints

Every stabilizer code satisfies

$$C_{ij} \equiv 0 \pmod{2},$$

since every pair of stabilizer generators commutes. Three possible constraints are:

- **Constant Local Non-Commutation:** Require

$$C_{ij} = \begin{cases} 0, & i = j, \\ 2a, & i \neq j, \end{cases}$$

where a is a positive integer satisfying $a < n$.

- **Quasi-Constant Local Non-Commutation:** Allow exactly two possible values:

$$C_{ij} = \begin{cases} 0, & i = j, \\ 2a \text{ or } 2b, & i \neq j, \end{cases}$$

where a and b are distinct positive integers satisfying $a, b < n$.

- **Low-Variance Local Non-Commutation:** Rather than prescribing exact values, require

$$\text{Var}(C_{ij}) \leq \sigma^2,$$

where σ is a prescribed positive constant.

Determining the search space retention factor for this constraint remains a nontrivial asymptotic enumeration problem, requiring constrained type enumeration or equivalent entropy-based methods.

3.7 Automorphism Groups

Automorphism groups provide a natural source of symmetry in combinatorial and algebraic structures. Many known semi-perfect quantum codes exhibit substantial symmetry, making prescribed automorphisms useful structural constraints that can reduce the search space by eliminating equivalent configurations and redundant constructions.

Possible restrictions include cyclic, dihedral, Mathieu, primitive, transitive, k -transitive, generator-transitive, or semiregular automorphism groups [8]. More generally, one may impose constraints on orbit structures or prescribed orbit decompositions, or require specified structural properties of the automorphism group without fixing the group a priori.

The corresponding search-space retention factor is generally case-dependent, as the reduction in the search space depends on the particular automorphism group and its action on the qudits and stabilizer generators. In particular, the retention factor may depend on the resulting orbit structure and orbit sizes, and therefore no universal expression is assigned to automorphism-group constraints. Its value can instead be evaluated for a specified group or orbit decomposition.

3.8 Mixed Constraints

When m constraints with individual search-space retention factors $\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_m$ are applied simultaneously to a code, the combined effective retention factor $\mathcal{R}_{\text{eff}}$ satisfies

$$\prod_{i=1}^m \mathcal{R}_i - d \leq \mathcal{R}_{\text{eff}} \leq \min_{1 \leq i \leq m} \mathcal{R}_i.$$

Here, d is the clearance constant, which quantifies the degree to which the m constraints clear the search space:

$$d = \min \left(\prod_{i=1}^m \mathcal{R}_{i, \log_{S_O}(S)} \right),$$

where

$$S = \frac{S_O^{\prod_{i=1}^m \mathcal{R}_i}}{\sum_{i=1}^m S_O^{\mathcal{R}_i} - (m-1)S_O}, \quad S \geq 1.$$

Two constraints are considered independent if applying one does not alter the retention factor of the other when enforced on the already reduced search space. Mathematically, independence requires:

$$\frac{\log S_{R_2}}{\log S_O} = \frac{\log S_{R_{\text{eff}}}}{\log S_{R_1}},$$

where S_{R_i} denotes the search-space size after imposing constraint i alone, and $S_{R_{\text{eff}}}$ denotes the size after imposing both constraints. Under strict independence, the effective retention factor reduces to:

$$\mathcal{R}_{\text{eff}} = \mathcal{R}_1 \mathcal{R}_2$$

More generally, if all m constraints are mutually independent,

$$\mathcal{R}_{\text{eff}} = \prod_{i=1}^m \mathcal{R}_i.$$

For other cases, determining the exact value of $\mathcal{R}_{\text{eff}}$ requires a case-wise analysis of the specific constraints.

4 Experimental Analysis

4.1 Asymptotic Behavior of the Correction Factor

Here we briefly state the behavior of the correction factor C for the weight-spectrum constraint; the same analysis can be generalized to other constraints.

4.1.1 Methodology

We consider known quantum codes and their corresponding weight spectra $\mathcal{W} = \{w_1, w_2, \dots, w_m\}$. The search-space retention factor for the weight-spectrum constraint is calculated as

$$\mathcal{R} = \frac{\log_{q^2} S_R}{\log_{q^2} S_O} = H_{q^2}(\delta_{\text{max}}) - C,$$

where

$$S_R = \left(\sum_{i=1}^m \binom{n}{w_i} (q^2 - 1)^{w_i} \right)^{n-k_n}, \quad \delta_{\text{max}} = \frac{\max(\mathcal{W})}{n}.$$

The corresponding correction factor C is evaluated for each code.

We examined 58,902 code parameter sets, of which 34,775 codes contained usable published matrix data, across the six field sizes $q = 2, 3, 4, 5, 7, 8$. The data were obtained from codetables.de [6].

4.1.2 Empirical Observation

The calculated correction factors exhibit a clear dependence on the block length n and the code dimension k . In particular, C decreases as either n or k increase, indicating that the finite-size correction becomes progressively smaller for longer codes. Figures 4.1.2 (a) and 4.1.2 (b) show this behavior for the qubit case ($q = 2$), while the corresponding results for the other field sizes are provided in the Appendix.

The observed decrease is consistent with the fact that the entropy term provides the leading extensive contribution to the logarithmic search-space size, whereas the correction arises from finite-size effects and from the presence of multiple weights in the spectrum.

4.1.3 Analytical Form of the Finite-Size Correction

The observed behavior can be further understood by expanding the finite-size expression using Stirling's approximation. Defining

$$X = \sum_{w \in \mathcal{W}} \binom{n}{w} (q^2 - 1)^w, \quad X_{\text{dominant}} = \binom{n}{w_{\text{max}}} (q^2 - 1)^{w_{\text{max}}},$$

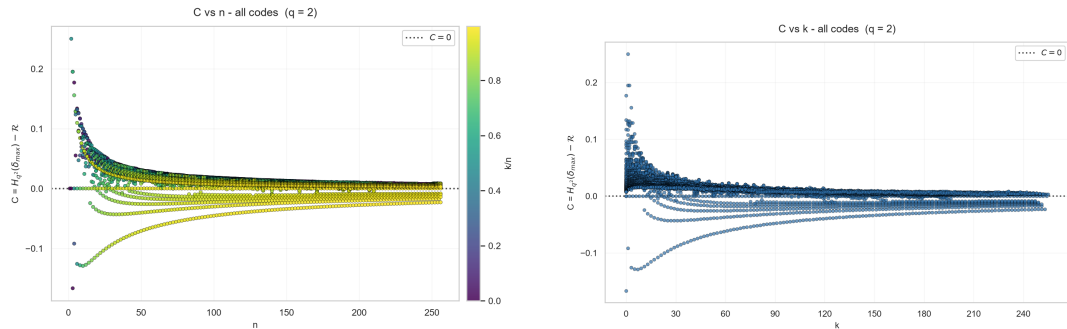
the correction factor becomes

$$C_n = \frac{\log_{q^2} [2\pi n \delta_{\text{max}} (1 - \delta_{\text{max}})]}{2n} - \frac{1}{n} \log_{q^2} \left(\frac{X}{X_{\text{dom}}} \right) + o\left(\frac{1}{n}\right).$$

Here, X represents the total combinatorial contribution of all distinct weights, while X_{dominant} represents the contribution of the maximum weight w_{max} . The first term captures the finite- n correction to the dominant contribution, while the second accounts for contributions from the remaining weights.

Thus, the observed behavior is consistent with a logarithmically corrected inverse-linear finite-size scaling,

$$C = O\left(\frac{\log n}{n}\right).$$



(a) Correction factor C versus block length n for the weight-spectrum constraint at $q = 2$ based on 17,882 data points.

(b) Correction factor C versus encoded qubits k for the weight-spectrum constraint at $q = 2$ based on 17,882 data points

Figure 4.1.2: Finite-size correction factor C for $q = 2$.

4.1.4 Empirical Scaling Across Field Sizes

To characterize the observed scaling more quantitatively, we fitted the empirical upper envelope of C separately for each field size to the form

$$C_{\max}(n) \approx \alpha n^{\beta}.$$

The resulting fits are shown below.

q	α	β	$R_{\log}^2$
2	0.614	-0.775	0.993
3	0.314	-0.715	0.994
4	0.271	-0.744	0.988
5	0.265	-0.784	0.981
7	0.190	-0.743	0.987
8	0.182	-0.749	0.984

Table 4.1.4: Power-law fits to the empirical upper envelope of the finite-size correction factor.

Although a power law fits the data well over the investigated range, the fitted exponents should be interpreted as effective finite-range exponents rather than evidence of a pure power-law asymptotic form. The analytical expression instead predicts logarithmically corrected inverse-linear scaling. Across field sizes, we find

$$\alpha \ln(q^2) = 0.774 \pm 0.060, \quad \beta = -0.752 \pm 0.023.$$

yielding the empirical envelope

$$C_{\max}(n) \approx \frac{0.77}{\ln(q^2)} n^{-0.75}$$

This provides a compact empirical description of the data, while the analytical expression gives the underlying finite-size interpretation.

4.1.5 Parameter-Free Upper Bound

The analytical expression also yields a parameter-free bound for the dominant finite-size contribution. Since

$$\delta_{\max}(1 - \delta_{\max}) \leq \frac{1}{4},$$

the dominant term satisfies

$$C_{\text{dominant}} \leq \frac{\log_{q^2}(\pi n/2)}{2n}.$$

The remaining non-dominant-weight contribution is represented explicitly in the preceding expression.

Empirically, this bound on C is satisfied by 99.0–99.7% of the codes for every investigated field size, with violations confined to small block lengths and none observed beyond $n = 118$. This agreement supports the predicted $n^{-1} \log n$ scaling of the dominant finite-size contribution, while the remaining non-dominant-weight contribution is treated separately.

4.1.6 Asymptotic Interpretation

The correction factor therefore decreases toward zero as n increases,

$$\lim_{n \rightarrow \infty} C = 0,$$

consistent with the asymptotic retention factor defined previously. The analytical form indicates logarithmically corrected inverse-linear scaling, while the non-dominant-weight contribution remains non-negligible over the investigated range, increasing from approximately 4% to 13% at $q = 2$

4.1.7 Entropy-Peak Structure of the Correction Factor

A second structural feature emerges from the dependence of C on $\delta_{\max}$. Since the q^2 -ary entropy is maximized at

$$\delta^* = 1 - \frac{1}{q^2}.$$

we find that, 100% of the codes with $C < 0$, for every investigated field, satisfy

$$\delta_{\max} > \delta^*.$$

Thus, within the investigated dataset, negative finite-size corrections occur exclusively beyond the entropy maximum. Beyond this point, the largest-weight contribution need not dominate the total combinatorial sum, making the detailed weight distribution increasingly important.

4.1.8 Predictability from Global Code Parameters

Finally, we examine whether C can be predicted from the global parameters (n, k, q) without detailed knowledge of the individual weight structure. Using a pooled linear model with finite-size features constructed from (n, k, q) , we find two distinct regimes separated by the entropy peak.

For

$$\delta_{\max} \leq 1 - \frac{1}{q^2},$$

the model achieves $R^2 \approx 0.948$, indicating substantial agreement between the fitted dependence and the observed correction factors. Beyond the entropy peak, this decreases sharply to $R^2 \approx 0.122$, indicating a stronger dependence on the individual code's weight spectrum and stabilizer structure.

Thus, within the investigated dataset, global parameters provide a useful estimate of C below the entropy peak, potentially allowing the search-space retention factor of unresolved codes to be estimated without detailed weight-spectrum information. Beyond the entropy peak, reliable estimation likely requires additional information about the individual code's weight structure.

4.2 Symmetrical Behavior of Semi-Perfect Quantum Codes

Due to their high structural symmetry, semi-perfect quantum codes possess a finite weight spectrum. Consequently, it is natural to replace the maximum weight $\max(\mathcal{W})$ with the average weight w_{avg} .

We define the normalized average relative weight δ_{avg} and the modified correction factor C' as

$$\delta_{\text{avg}} = \frac{w_{\text{avg}}}{n}, \quad C' = H_{q^2}(\delta_{\text{avg}}) - \mathcal{R}.$$

4.2.1 Empirical Asymptotic Behavior of C'

Analyzing all 1,909 known semi-perfect quantum codes across $q \in \{2, 3, 4, 5, 7, 8\}$, we observe that semi-perfect quantum codes satisfy

$$\lim_{n \rightarrow \infty} C' = 0.$$

This asymptotic convergence is specific to the semi-perfect codes considered here and is not observed for general quantum codes.

The corresponding dependence of C' on n and k is shown in Figures 4.2.1 (a) and 4.2.1 (b), respectively, for semi-perfect quantum codes at $q = 2$. Results for the remaining field sizes, as well as for all codes, are provided in the Appendix.

To characterize the empirical asymptotic envelope, we define

$$C'_{\max}(n) = \max_k C'(n, k).$$

The upper envelope is empirically well described by

$$C'_{\max}(n) \approx \frac{b_q}{n}.$$

The fitted parameters are shown in Table 4.2.1.

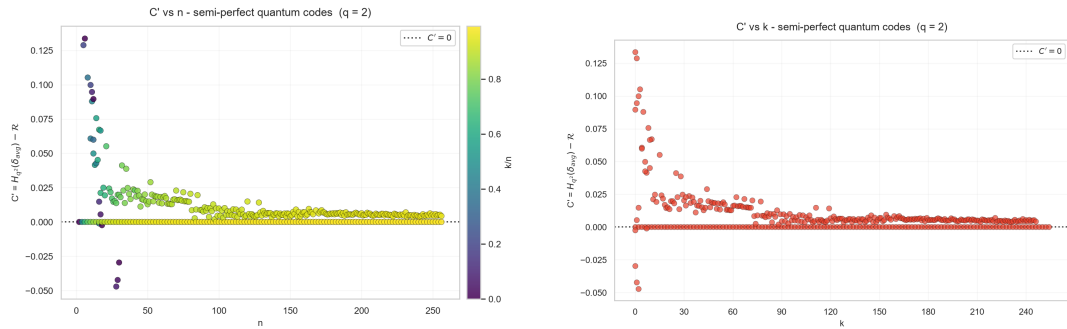
q	b_q	R^2
2	0.9634	0.8353
3	0.5161	0.7455
4	0.2584	0.7935
5	0.2326	0.7601
7	0.2628	0.8430
8	0.3046	0.8459

Table 4.2.1: Empirical fit coefficients for the upper envelope $C'_{\max}(n) \approx b_q/n$.

The fitted envelopes establish an empirical decay scale

$$C' = O\left(\frac{1}{n}\right).$$

Thus, the modified correction factor exhibits an empirical $\frac{1}{n}$ decay, in contrast to the $\frac{\log n}{n}$ decay observed for the standard maximum-weight formulation.



(a) Modified correction factor C' versus block length n for the weight-spectrum constraint for semi-perfect quantum codes at $q = 2$ based on 384 data points.

(b) Modified correction factor C' versus encoded qubits k for the weight-spectrum constraint for semi-perfect quantum codes at $q = 2$ based on 384 data points

Figure 4.2.1: Finite-size correction factor C' at $q = 2$.

4.2.2 Comparison of C' and C

We further compare C' with the original correction factor C for the 1,909 semi-perfect quantum codes. The difference satisfies

$$C' - C = H_{q^2}(\delta_{\text{avg}}) - H_{q^2}(\delta_{\text{max}}),$$

with numerical agreement to machine precision. Since $\delta_{\text{avg}} \leq \delta_{\text{max}}$ and H_{q^2} increases up to $\delta^* = 1 - q^{-2}$ before decreasing, the sign of $C' - C$ is determined by the positions of δ_{avg} and δ_{max} relative to the entropy maximum.

For 1,289 codes (67.52%), $C' - C = 0$, while $C' - C > 0$ and $C' - C < 0$ occur for 344 (18.01%) and 276 (14.45%) codes, respectively. The magnitude of $C' - C$ generally decreases with increasing n , with a pooled Spearman correlation of $\rho = -0.43$. Its magnitude is small, ranging from -0.0919 to 0.0280 , while the fraction satisfying $C' = C$ increases from 38.5% at $q = 2$ to 92.3% at $q = 8$.

This indicates that the average-weight and maximum-weight formulations become increasingly similar for longer semi-perfect quantum codes.

4.2.3 Implications for Unresolved Semi-Perfect Quantum Codes

Using C' , we construct empirical estimates of the average weight and full weight spectrum, including frequencies, for unresolved semi-perfect quantum codes. These estimates reduce the raw search space by a factor of $10^{8.3608}$ to $10^{193.9298}$, corresponding to an equivalent runtime speedup in code-search procedures.

Code	Weight Spectrum (without frequencies)	$\mathcal{R}$	C'	w_{avg}	Weight Spectrum (with frequencies)	Runtime Speedup
[[14, 3, 5]]	{8, 10}	0.9368	0.0530 ± 0.006	9.4582 ± 0.3464	{8:3, 10:8}	$10^{8.6602}$
[[17, 3, 5]]	{10, 12}	0.9425	0.0364 ± 0.006	10.9035 ± 0.2845	{10:8, 12:6}	$10^{13.7075}$
[[27, 18, 4]]	{18, 20, 22}	0.9763	0.0151 ± 0.006	18.4165 ± 0.7502	{18:6, 20:2, 22:1}	$10^{8.3608}$
[[73, 18, 20]]	{40, 42}	0.9260	0.0181 ± 0.006	41.5586 ± 0.5328	{40:15, 42:40}	$10^{193.9298}$

Table 4.2.3: Empirical parameter estimates and runtime speedups for unresolved semi-perfect quantum codes.

The weight spectra without frequencies and the C' values used to derive $\mathcal{R}$ were estimated empirically and verified manually. The weighted spectra with frequencies were manually selected to match the empirically derived average weight w_{avg} , and the corresponding runtime speedups were calculated using these weighted spectra.

The codes shown above are representative examples; the same procedure can be applied to any unresolved semi-perfect quantum code to empirically estimate its weight spectrum, including the corresponding frequencies.

4.3 Runtime Analysis

We evaluate runtime performance using the $[[5, 1, 3]]$ code to assess the relationship between the search-space retention factor and computational cost. For a search-space retention factor $\mathcal{R}$, the theoretical runtime speedup relative to the unconstrained search space S_O is

$$S_O^{1-\mathcal{R}}.$$

To establish a strict upper bound, we benchmark the relative search duration over the full reduced search space without early termination, corresponding to the worst-case scenario in which only a single valid stabilizer configuration exists.

The empirical efficiency ratio is defined as

$$\eta = \frac{\text{Experimental Runtime Speedup}}{\text{Theoretical Runtime Speedup}} = \frac{T_O/T_R}{S_O^{1-\mathcal{R}}},$$

Constraint	Search-Space Retention Factor (R)	Runtime (s)	Experimental Speedup (T_O/T_R)	Theoretical Speedup (S_O^{1-R})	Empirical Efficiency Ratio (η)
None (Unconstrained)	1.00	13,851.05	1.00 (Baseline)	1.00	1.00
Universal stabilizer constraints	0.8469	342.99	40.3832	69.6902	0.5794
Weight spectrum $\mathcal{W} = \{4\}$	0.8661	361.91	38.2720	40.8676	0.9477
Cyclicity	0.2500	0.000227	$10^7.7854$	2^{30}	0.0568

Table 4.3: Runtime analysis and empirical performance metrics for different search constraints applied to the $[[5, 1, 3]]$ code.

and quantifies the agreement between the experimental and theoretical speedups. The observed discrepancies arise because the theoretical model does not account for the computational cost of evaluating constraints. Universal stabilizer constraints yield $\eta = 0.5794$, reflecting the overhead of commutation testing and rank maintenance. Cyclicity yields a lower efficiency ratio of $\eta = 0.0568$, as the extremely short runtime makes fixed validation overhead comparatively significant.

As the search space grows, the theoretical speedup increases exponentially with $n(n - k)$, whereas constraint-validation costs per search node remain approximately fixed or grow more slowly. Consequently, the empirical efficiency ratio may decrease for stronger constraints and larger search spaces. However, because the magnitude of the theoretical runtime speedup becomes extraordinarily large for vast search spaces, this reduction in empirical efficiency has a negligible impact on the net performance gains. Furthermore, optimized software implementations that streamline constraint validation logic can significantly reduce this overhead and improve overall execution efficiency.

5 Application to the $[[14, 3, 5]]_2$ Quantum Code

We apply the proposed framework to construct the best-known quantum code with parameters $(n, k, q) = (14, 3, 2)$. Although the resulting code contains a small number of residual undetected errors, the resulting code effectively functions as a practical alternative to the theoretically optimal $[[14, 3, 5]]_2$ code. The methodology provided here is general and extends to other quantum code search problems.

Section 4.2 provides an empirical method to estimate the average weight of unresolved semi-perfect quantum codes. Given that the optimal $[[14, 3, 5]]_2$ code follows the definition of semi-perfect quantum codes, its average generator weight is estimated at 9.4582 ± 0.3464 . This corresponds to candidate weight-spectrum frequencies for $\{8, 10\}$ of $\{1, 10\}$, $\{2, 9\}$, $\{3, 8\}$, or $\{4, 7\}$. By analyzing neighboring codes and the expected average weight, $\{8 : 3, 10 : 8\}$ is selected as the most accurate empirical weight spectrum for the $[[14, 3, 5]]_2$ code.

5.1 Methodology

To make the search space tractable, we decompose the generation process into $|\mathcal{W}|$ sequential stages:

1. **Stage 1:** Identify inequivalent classes of the first f_1 generators with weight w_1 that maximize the detection of errors with weight $w < d_{\max}$.
2. **Subsequent Stages:** Determine inequivalent classes for f_i generators of weight w_i that maximize error detection over the remaining undetected errors from preceding stages.

Because semi-perfect quantum codes exhibit structural symmetry, their weight spectra vary minimally, typically requiring only 2 or 3 search stages.

To accelerate computation across stages:

- **Stage 1 Pruning:** Local-Clifford and permutation symmetry reductions, combined with an exact Walsh-transform objective and admissible branch-and-bound pruning, reduce the Stage 1 search space drastically.
- **Stage 2+ Pruning:** Centralizer and quotient-space reduction, exact Walsh–Hadamard objective evaluation, Gray-code candidate enumeration, and greedy/beam/LNS search over true error coverage reduce the complexity of subsequent stages, making otherwise intractable searches feasible.

5.2 Results for (14, 3, 2) Parameters

Stage 1 was exhaustively solved in 0.26 seconds using 3 weight-8 generators. The optimization yielded 148 distinct optimal column multisets modulo permutation, which reduced under generator permutations to 45 inequivalent Stage 1 orbits. Every orbit achieved the proven Stage 1 optimum, detecting 80,584 of the 91,770 total errors of weight $w \leq 4$ and leaving 11,186 errors undetected.

A complete sweep over all 45 Stage 1 orbits for Stage 2 completed in 19.4 minutes. Across all orbits, Stage 2 configurations detected between 11,175 and 11,179 of the remaining Stage 1 errors:

- **7 orbits** yielded 7 residual errors each, detecting

$$\frac{91,763}{91,770} \approx 99.9923\%$$

of errors.

- The 7 optimal orbits exhibited 3 distinct weight profiles for their residual errors (weights 1 / 2 / 3 / 4):

1. **4 orbits:** Profile 3/3/1/0, with errors generated by X_a, X_b, X_c for

$$(a, b, c) \in \{(1, 3, 4), (1, 3, 7), (2, 3, 4), (2, 5, 9)\}.$$

2. **2 orbits:** Profile 1/3/3/0, with errors generated by $X_a, P_b P_c$, $P \in \{X, Z\}$, for

$$(a, b, c) \in \{(1, 2, 5), (1, 2, 7)\}.$$

3. **1 orbit:** Profile 1/1/2/3, exhibiting low error correlation.

A deeper search pass executed on these 7 candidate orbits led to an improved result for the orbit with the 1/1/2/3 residual profile at $t = 1,800.23$ seconds. The refined search yielded a residual error weight profile of **0/0/2/3** (5 total residual errors), representing the best-known $[[14, 3]]_2$ code to date.

5.2.1 Best-Known Code Generators

Stage-1 Generators:	$g_1 = \text{IIIIXXXXIIZZXX}$
	$g_2 = \text{IIIIXXIIXXXXZZ}$
	$g_3 = \text{IIIIIZZZZZZZZZ}$
Stage-2 Generators:	$h_1 = \text{IXIXXXIZYZIXXX}$
	$h_2 = \text{IZXIXXYIXYIYXX}$
	$h_3 = \text{IIIZYXZXZXZIZY}$
	$h_4 = \text{IIYXZYIYZZYZX}$
	$h_5 = \text{IIZIXXZYXZYIZ}$
	$h_6 = \text{XIIIXYXZXIYIXX}$
	$h_7 = \text{IIYIZXZXYYXIYX}$
	$h_8 = \text{ZIIIZXIZZYZZYXX}$

5.2.2 Residual Undetected Errors

The 5 remaining undetected errors span all physical qubits and are not strictly X -type:

- **Weight 3 (2 errors):**
 - $\text{IIXIIIIIIIZZ} \equiv X_4 Z_{13} Z_{14}$
 - $\text{IIIIIXZIIIXII} \equiv X_7 Z_8 X_{12}$
- **Weight 4 (3 errors):**
 - $\text{YIIIZIIHYIXII} \equiv Y_1 Z_6 Y_{10} X_{12}$
 - $\text{IXIIIZYXIIII} \equiv X_2 Z_7 Y_8 X_9$
 - $\text{IIZIXIIIIIXII} \equiv Z_3 X_5 X_{11} X_{12}$

5.3 Error Correction and Detection Performance

The constructed code detects

$$\frac{91,765}{91,770} \approx 99.9945\%$$

of errors of weight $w \leq 4$. Furthermore, it produces distinct syndromes for

$$\frac{808}{862} \approx 93.7354\%$$

of errors with weight $w \leq 2$. The remaining 54 errors form 27 degenerate pairs with matching syndromes; for each pair, a correction strategy can be defined to either correct the error outright or map it to one of the 5 residual undetected errors. As a result, the code corrects

$$\frac{835}{862} \approx 96.8677\%$$

of errors with $w \leq 2$.

5.3.1 Canonical Representation

The canonical generator matrix representation for this code is

$$\begin{pmatrix} \text{XZIZIIZIYIZZIY} \\ \text{IXZIZIZZYZIYY} \\ \text{IIXZIZIZYIZIZY} \\ \text{IZZIXIZIYIIXII} \\ \text{IIZIXZIZYIXIYX} \\ \text{IZZIZXIIZZYXYX} \\ \text{IZIZIZXIXIXIYZ} \\ \text{IIIZZIIZYIZXYZY} \\ \text{IZIIZZIZIYYIZZ} \\ \text{ZIZIZZIIZZIZZ} \\ \text{IIIIIZZZZZZZZ} \end{pmatrix}$$

5.4 Performance Under Stochastic Noise

Under an independent stochastic noise model where each qubit suffers an error with probability p , the logical success probabilities are derived as follows.

- **Probability of correcting errors with $w \leq 2$:**

$$P_{\text{correct}}(p) = \frac{(1-p)^{14} + 14p(1-p)^{13} + 88p^2(1-p)^{12}}{(1-p)^{14} + 14p(1-p)^{13} + 91p^2(1-p)^{12}}.$$

- **Probability of detecting errors with $w \leq 4$:**

$$P_{\text{detect}}(p) = \frac{(1-p)^{14} + 14p(1-p)^{13} + 91p^2(1-p)^{12} + \frac{9,826}{27}p^3(1-p)^{11} + \frac{81,078}{81}p^4(1-p)^{10}}{(1-p)^{14} + 14p(1-p)^{13} + 91p^2(1-p)^{12} + 364p^3(1-p)^{11} + 1,001p^4(1-p)^{10}}.$$

The corresponding probabilities are shown in Figure 5.4.

In physical hardware environments ($p \approx 10^{-2}$ to 10^{-3}), evaluating these expressions at $p = 10^{-2}$ yields

$$P_{\text{correct}}(10^{-2}) \approx 0.999733,$$

and

$$P_{\text{detect}}(10^{-2}) \approx 0.999999933.$$

These values confirm that the constructed code effectively performs as a theoretical $[[14, 3, 5]]_2$ code under realistic noise regimes.

6 Search Methodologies

The structural constraints developed in this work can be incorporated directly into computational searches rather than applied only as post-selection criteria. Depending on the implementation, stabilizer generators may be represented using either the binary symplectic or $GF(4)$ representation. The search may then proceed through either support-first construction, in which generator supports are selected before Pauli assignments, or direct construction, in which support and Pauli assignments are determined simultaneously. In the support-first approach, design-theoretic constraints can further restrict the search to structured support configurations arising from combinatorial objects such as balanced incomplete block designs (BIBDs), quasi-symmetric designs, finite geometries [9], and related incidence structures [10].

During candidate construction, the proposed constraints can be incorporated into backtracking [11], branch-and-bound [12], SAT/SMT [13], or integer-linear-programming (ILP) [14]

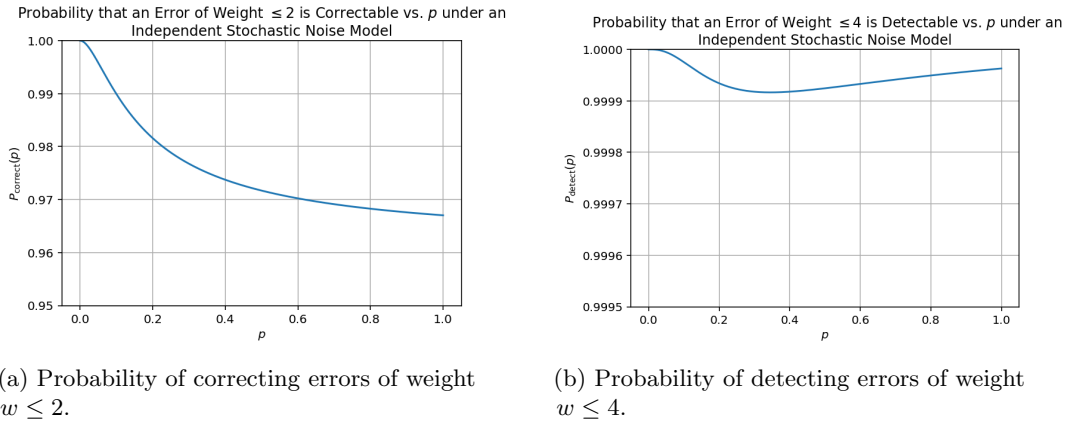


Figure 5.4: Error correction and detection probabilities as functions of the physical error rate p under an independent stochastic noise model.

searches. Constraints may be evaluated incrementally as generators are added, allowing candidates that violate weight, commutation, rank, distance, overlap, or other structural conditions to be discarded before completion. In this way, the theoretical search-space reduction quantified by $\mathcal{R}$ can be realized directly during the search rather than after candidate generation.

Symmetry-based constraints can be incorporated through canonicalization and isomorphism rejection [15], eliminating candidates equivalent under qudit or generator permutations. When a prescribed automorphism group or other symmetry is known, its orbit structure can similarly be used to restrict the search to inequivalent configurations. These methods are particularly relevant for semi-perfect codes, where the structural regularities investigated in Section 4 can provide additional restrictions.

Multiple constraints can be enforced simultaneously during candidate construction. When constraints are applied sequentially, candidates violating any previously imposed condition can be discarded before subsequent constraints are evaluated, while compatible constraints can be incorporated jointly into the same search formulation. The resulting reduction depends on the interaction between the constraints and need not equal the product of their individual reductions.

7 Discussion and Future Work

The results demonstrate that structural properties of quantum stabilizer codes can be converted into quantitative constraints on systematic code searches. The derived retention factors provide a common measure for comparing constraints, while the empirical correction analysis accounts for finite-length deviations from their asymptotic behavior. The analysis of semi-perfect codes further shows that structural information can be used to estimate search-space reductions even when complete weight spectra are unavailable, with the resulting estimates producing substantial computational speedups for unresolved codes. Runtime analysis confirms that these reductions can translate into practical computational savings, while the $(14, 3)$ qubit search demonstrates their utility in an actual high-distance code search.

Several directions remain for future work. The framework can be extended to additional structural constraints and larger qudit dimensions, while improved empirical models for finite-size corrections may further increase the accuracy of search-space estimates. Future searches can also integrate the framework directly with more sophisticated optimization, SAT/SMT, and symmetry-reduction methods. Finally, applying the framework to larger unresolved parameter sets may enable the systematic discovery of high-distance quantum stabilizer codes that are currently computationally inaccessible.

Author Contributions

U.K. conceived the study, developed the methodology, performed the analysis, and wrote the manuscript.

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None.

Competing Interests

The author declares no competing interests.

Code and Data Availability

The code used to perform the computations and generate the results presented in this work is publicly available on GitHub ([utsavkaria-tech](https://github.com/utsavkaria-tech)). The data generated during the study are available from the corresponding code repository.

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Appendix A: Additional Experimental Results

This appendix presents additional experimental results supporting the analyses in Section 4. In particular, we provide the corresponding results for field sizes $q = 3, 4, 5, 7, 8$, complementing the $q = 2$ results presented in the main text.

A.1 - Correction Factor C

The dependence of the finite-size correction factor C on the block length n and encoded qubits k is shown for $q = 3, 4, 5, 7, 8$. These results complement the corresponding $q = 2$ results presented in Section 4.1.

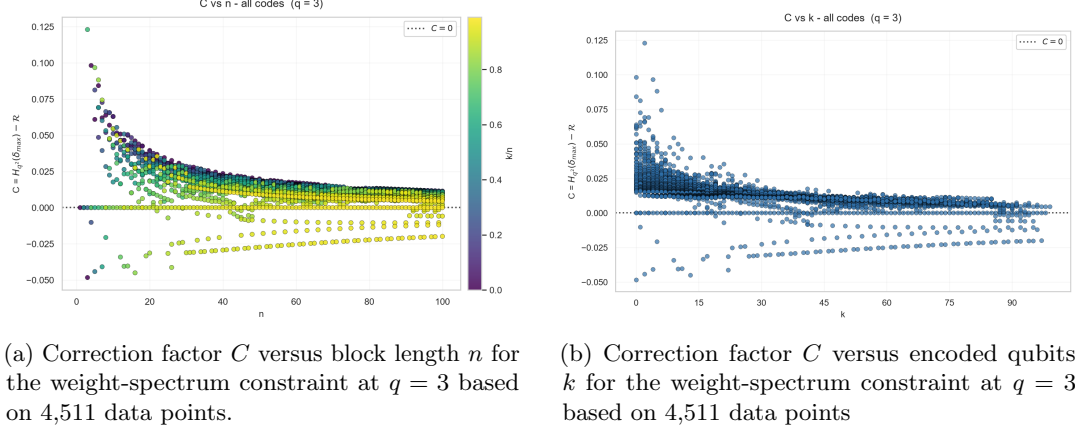


Figure A.1-1: Finite-size correction factor C for $q = 3$.

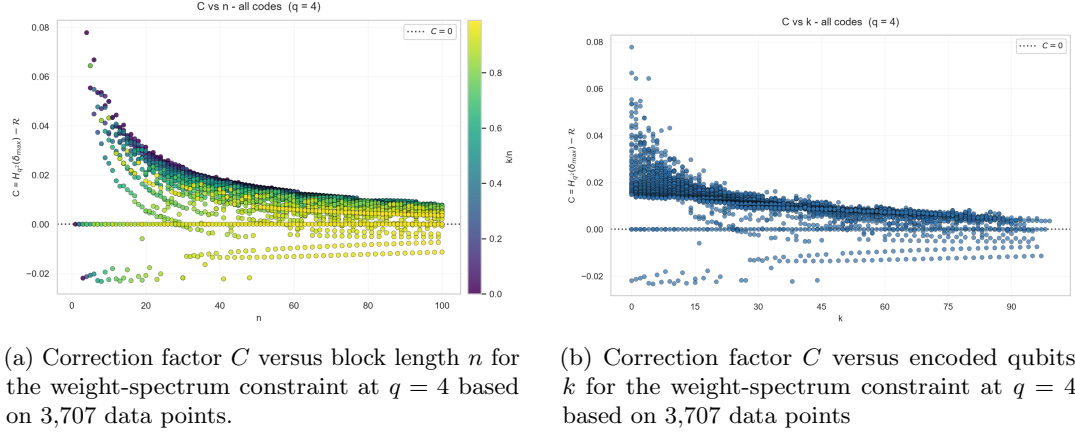


Figure A.1-2: Finite-size correction factor C for $q = 4$.

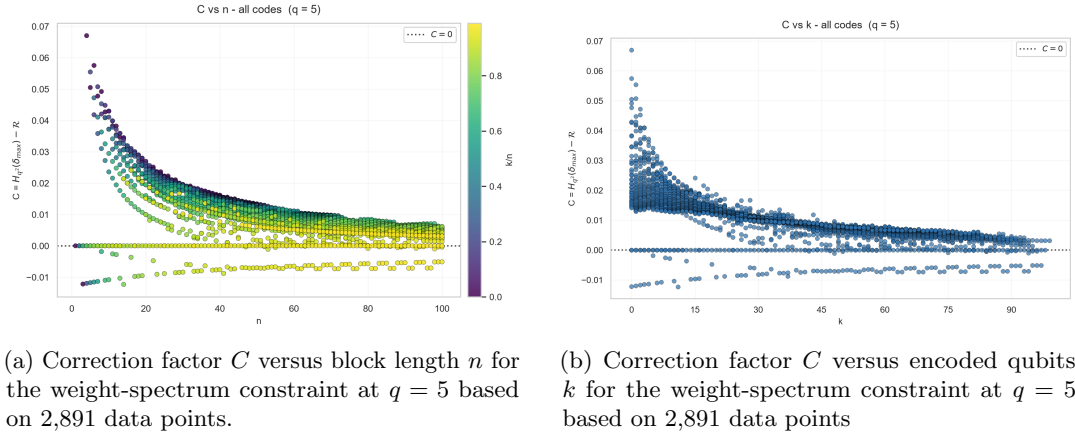


Figure A.1-3: Finite-size correction factor C for $q = 5$.

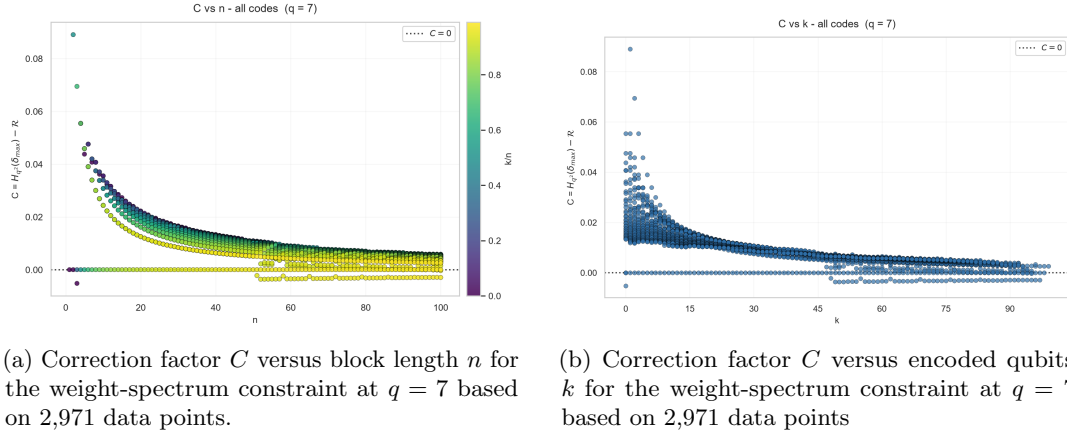


Figure A.1-4: Finite-size correction factor C for $q = 7$.

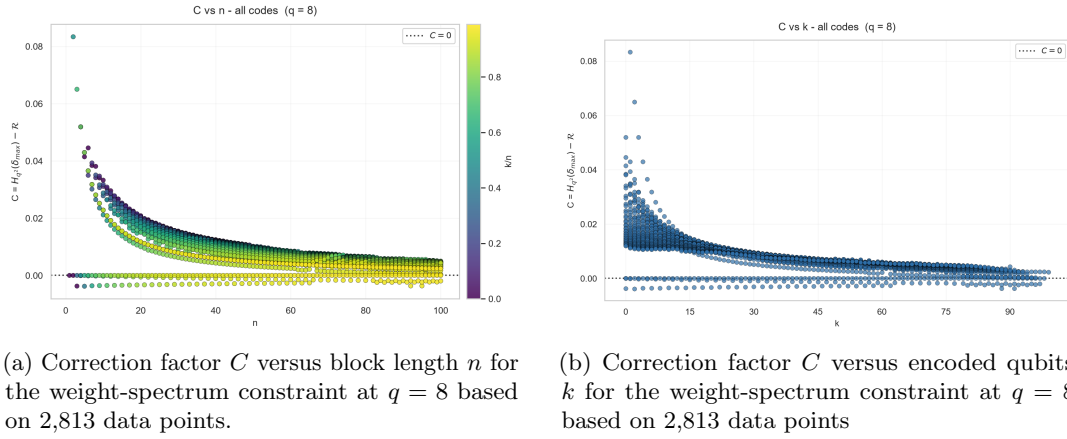


Figure A.1-5: Finite-size correction factor C for $q = 8$.

A.2 - Modified Correction Factor C'

The corresponding dependence of the modified correction factor C' on n and k is shown for $q = 3, 4, 5, 7, 8$, providing additional evidence for the empirical asymptotic behavior discussed in Section 4.2.

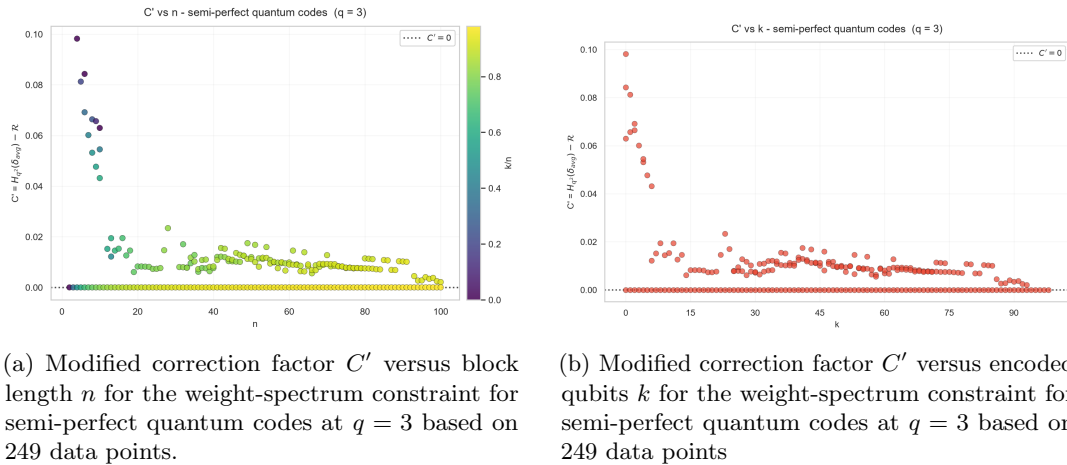
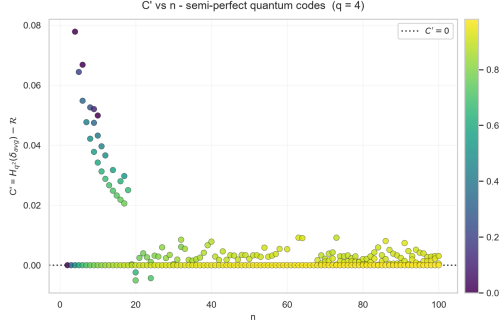
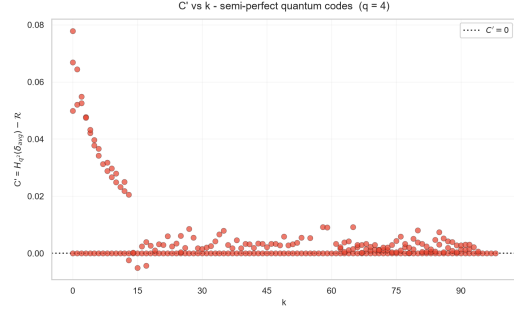


Figure A.2-1: Finite-size correction factor C' at $q = 3$.

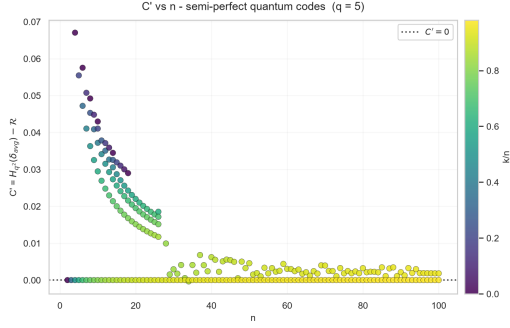


(a) Modified correction factor C' versus block length n for the weight-spectrum constraint for semi-perfect quantum codes at $q = 4$ based on 236 data points.

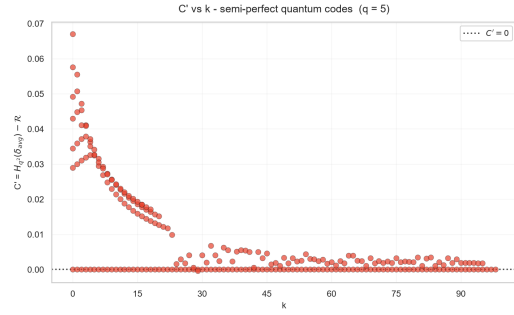


(b) Modified correction factor C' versus encoded qubits k for the weight-spectrum constraint for semi-perfect quantum codes at $q = 4$ based on 236 data points

Figure A.2-2: Finite-size correction factor C' at $q = 4$.

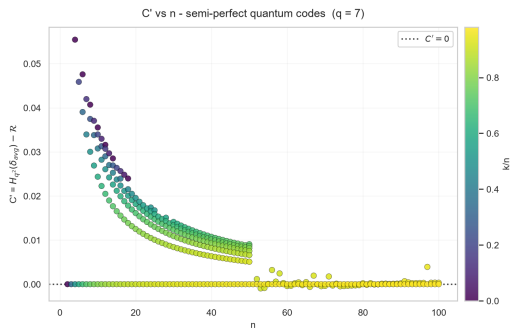


(a) Modified correction factor C' versus block length n for the weight-spectrum constraint for semi-perfect quantum codes at $q = 5$ based on 245 data points.

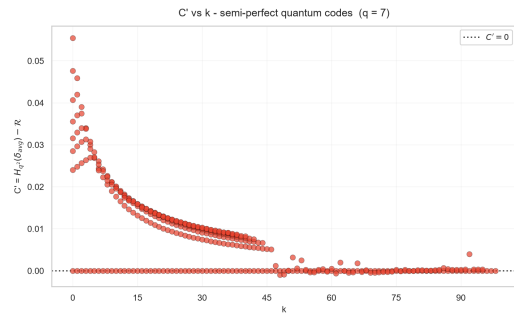


(b) Modified correction factor C' versus encoded qubits k for the weight-spectrum constraint for semi-perfect quantum codes at $q = 5$ based on 245 data points

Figure A.2-3: Finite-size correction factor C' at $q = 5$.

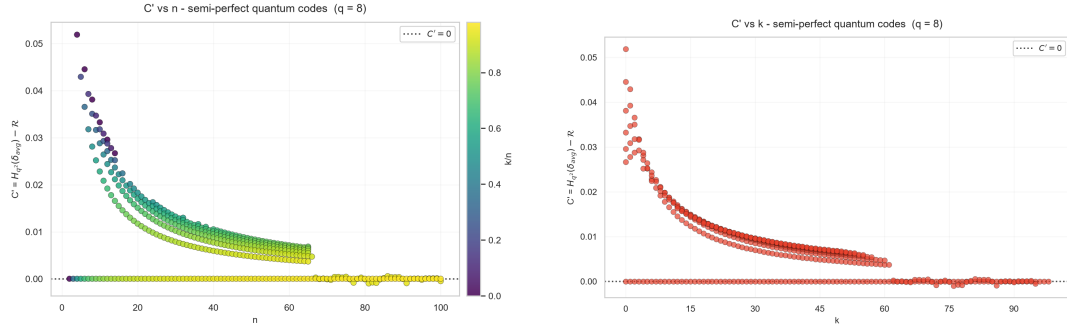


(a) Modified correction factor C' versus block length n for the weight-spectrum constraint for semi-perfect quantum codes at $q = 7$ based on 353 data points.



(b) Modified correction factor C' versus encoded qubits k for the weight-spectrum constraint for semi-perfect quantum codes at $q = 7$ based on 353 data points

Figure A.2-4: Finite-size correction factor C' at $q = 7$.



(a) Modified correction factor C' versus block length n for the weight-spectrum constraint for semi-perfect quantum codes at $q = 8$ based on 442 data points.

(b) Modified correction factor C' versus encoded qubits k for the weight-spectrum constraint for semi-perfect quantum codes at $q = 8$ based on 442 data points

Figure A.2-5: Finite-size correction factor C' at $q = 8$.

A.3 - Additional Supporting Results

To further examine the analytical form of the finite-size correction, Figure A.3-1 compares the observed correction factor C with its single-dominant-term prediction across all investigated field sizes.

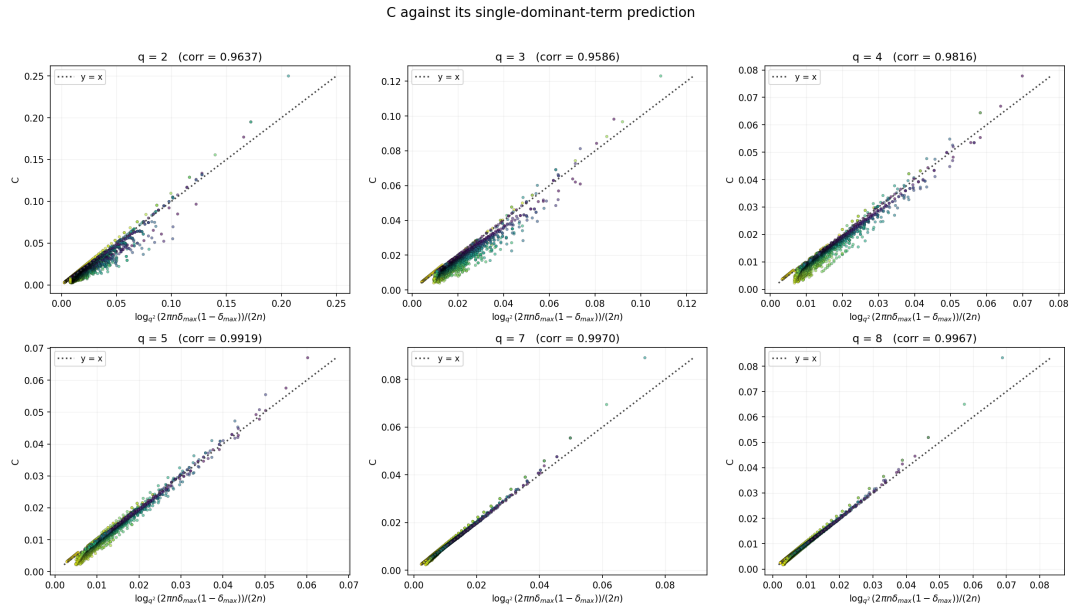


Figure A.3-1: Comparison of the finite-size correction factor C with its single-dominant-term prediction across field sizes $q = 2, 3, 4, 5, 7, 8$.

The close agreement across the investigated field sizes provides additional support for the analytical finite-size correction derived in Section 4.1.

Additional plots showing the dependence of C' on n for all codes, with semi-perfect quantum codes highlighted, are also provided to further illustrate the asymptotic behavior discussed in Section 4.2.

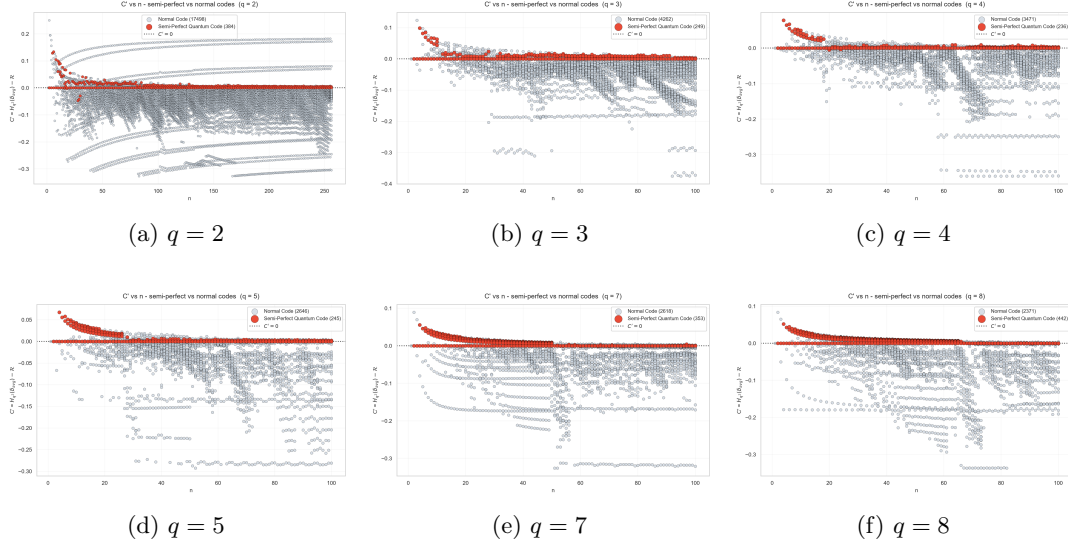


Figure A.3-2: Modified correction factor C' versus block length n for all investigated quantum codes, with semi-perfect quantum codes highlighted, across field sizes $q = 2, 3, 4, 5, 7, 8$.

The plots below show the relationship between $H_{q^2}(\delta_{\max})$ and $\mathcal{R}$ across the full dataset for $q = 2, 3, 4, 5, 7, 8$. The results provide additional characterization of the maximum-weight formulation and its dependence on the search-space retention factor across different field sizes.

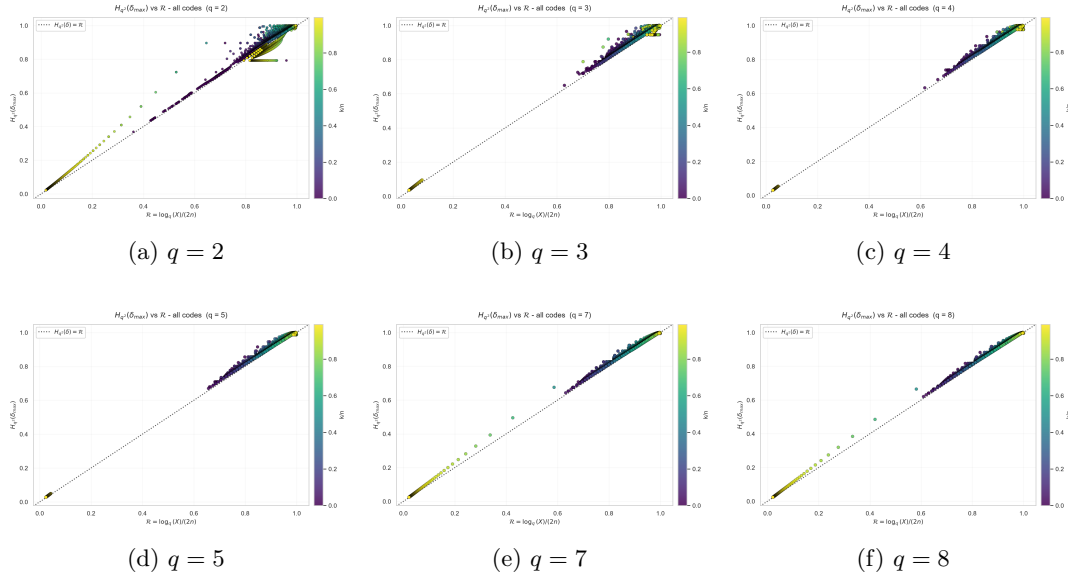


Figure A.3-3: The q^2 -ary entropy $H_{q^2}(\delta_{\max})$ versus the code rate R for all investigated quantum codes across field sizes $q = 2, 3, 4, 5, 7, 8$.